

DIABETES RISK PREDICTION: USING MACHINE LEARNING

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1. INTRODUCTION

Project Objective

- Develop a predictive machine learning pipeline for healthcare clinics.
- Classify patients as **high-risk or low-risk** for diabetes.
- Support early clinical intervention and preventive care.

Dataset Overview

- **Total Records:** 768 patients.
- **Total Features:** 8 biometric markers, 1 target column.
- **Target Mapping:** 0 = No Diabetes, 1 = Has Diabetes.

Target Variable Distribution

- **Dataset baseline:** 768 total patient records.
- **Class 0 (Healthy):** 500 patients represent 65.10% of data.
- **Class 1 (Diabetic):** 268 patients represent 34.90% of data.
- **Distribution state:** Significant class imbalance exists.
- **Ratio impact:** Healthy cases nearly double the diabetic cases.

2. DATA CLEANING & PREPROCESSING

Missing Data Audit

- **Glucose:** 5 missing entries.
- **BloodPressure:** 35 missing entries.
- **SkinThickness:** 227 missing entries.
- **Insulin:** 374 missing entries.
- **BMI:** 11 missing entries.

Cleaning Methodology

- **Action:** Applied Median Imputation across all structural data gaps.
- **Justification:** Row deletion would discard roughly 48% of the patient database.
- **Strategy:** Median selected over mean to negate extreme biometric outliers (e.g., heavily skewed Insulin distributions).
- **Result:** Successfully restored dataset to 768 complete, non-null patient profiles.

3. ANALYSIS & VISUAL INSIGHTS

Fig 1: Outcome Baseline Distribution

- **Observation:** The sample heavily favors non-diabetic records.
- **Insight:** Predictive modeling must account for this 65/35 imbalance to avoid over-predicting the majority class.

Fig 2: Glucose Spread Comparison

- **Non-Diabetic Median:** 107.5 mg/dL.
- **Diabetic Median:** 140.0 mg/dL.
- **Insight:** Blood sugar is a high-yield diagnostic baseline marker showing clear separation between classes.

Fig 3: Feature Correlation Matrix

- **Glucose Connection:** Strongest positive linear link to target column (0.49).
- **BMI Connection:** Secondary positive linear risk driver (0.31).
- **Age Connection:** Tertiary risk modifier (0.24).
- **BloodPressure Connection:** Weakest observed statistical dependency (0.16).
- **Insight:** Elevated Glucose and high BMI represent the primary risk engines within this cohort.

4. MODEL DEVELOPMENT & EVALUATION

Framework Setup

- **Algorithm:** Baseline Logistic Regression (max iterations = 1000).
- **Data Partitioning:** 80% Training, 20% Testing.
- **Validation Control:** Stratified split applied to lock matching target distributions.

Performance Evaluation Metrics

- **Accuracy Score (0.7013):** Model correctly identifies 70.13% of all testing scenarios.
- **Precision Score (0.5870):** Flags positive risks with a 58.70% exact success rate.
- **Recall Score (0.5000):** Captures exactly 50.00% of all real diabetic individuals.
- **F1 Score (0.5400):** Harmonic compromise balance of the system.

Confusion Matrix Structural Breakdown

Confusion Matrix Layout:

- **Accurate Healthy Identifications:** The model correctly confirmed 81 healthy patients as negative (True Negatives).
- **Critical Diagnostic Omissions:** The system completely missed 27 diabetic patients, posing high medical risk (False Negatives).
- **Successful Diabetic Detections:** The model accurately caught and flagged 27 active diabetic cases (True Positives).
- **Unnecessary Secondary Alerts:** The system generated 19 false alarms for healthy patients (False Positives).

Medical Context Interpretation

- **The Clinical Issue:** Medical screeners require high Recall over Precision.
- **The Reality:** A false alert triggers a harmless second lab test.
- **The Danger:** A false negative sends a sick patient home completely unmonitored.
- **Verdict:** A 50.00% Recall rate is unsafe for unsupervised clinic deployment.

5. CONCLUSION & RECOMMENDATIONS

Capstone Project Summary

- Successfully engineered an end-to-end medical prediction workflow.
- Resolved data gaps using robust median-based cleaning strategies.
- Tested an interpretable linear algorithm to establish performance thresholds.

Deployment Strategy

- Use strictly as a passive pre-screening filter inside patient check-in portals.
- Flags high-risk patient patterns instantly for immediate manual physician review.
- Never use as a standalone or automated diagnostic tool.

Roadmap For Model Optimisation

1. **Feature Scaling:** Implement StandardScaler to compress varying parameter ranges (e.g., Insulin vs Age) for better weight assignment.
2. **Feature Engineering:** Create custom hazard indicators compounding high BMI and high Glucose directly.
3. **Advanced Algorithms:** Pivot to non-linear ensemble models like Random Forest or XGBoost to capture complex biological interactions and push the Recall score safely past 85%.